



# New Draft Proposal

Challenge: Human Survivability

## PROPOSAL TITLE \* ()

Please provide the title of the technical solution you are proposing.

Hybrid propulsion system VTOL drone with tilt rotor system for medical and defence applications capable of long-range deliveries



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DynoBay

## TRL LEVEL \* ()

Please select the Technology Readiness Level ("TRL") of the technical solution you are proposing to the challenge.

TRL 4 Technology validated in lab

## SYSTEM TYPE \* ()

State whether your technical solution is software, hardware or a combination of both.

Both

## SYSTEM LEVEL \* ()

State whether your technical solution is by itself a platform or system, or whether it is a sub-system or component that would integrate into another system or platform.

System

## Short form proposal

### PART 1 - ALIGNMENT TO CHALLENGE \* ()

- What elements of the challenge are you addressing?
- How will your technical solution address the challenge?
- What is novel and or/unique about your solution?
- What defence and security needs are addressed by your technical solution?

-Addresses survivability gaps when evacuation is delayed, constrained, or denied.  
-Supports forces operating in contested ground and

### PART 2 - TECHNICAL APPROACH \* ()

- Describe your technical solution. You may attach graphics or images to help convey your concept.



airspace.

-Reduces dependence on ground convoys, helicopters, runways, and exposed personnel.

-Helps relieve pressure on overloaded local and rear medical capacity.

-Enables rapid movement of medical and survivability payloads to dispersed units.

-Supports prolonged field care by keeping casualties supplied until evacuation is possible.

-Provides VTOL access for austere locations and fixed-wing efficiency for useful range.

-Creates a dual-use capability for defence medical logistics, civil protection, disaster response, and emergency medical delivery.

### **PART 3 - TECHNOLOGY DEVELOPMENT ROADMAP WITH RISKS AND MITIGATIONS \* ()**

- How will the challenge funding help you to mature your proposed technical solution into a demonstrable and/or deployable technology?
- Key milestones:
  - Six months: What do you plan to achieve in six months?
  - One year: If selected for Mission Track and further challenge funding were offered, what would you intend to achieve in one year?
- Risks: What are the most significant risks associated with achieving these milestones?
- Mitigation: What measures are you proposing to mitigate these risks?

6 months: prove the operational feasibility of the prototype using segmented mission profiles

12m: Full scale solution development including all the missing scale-related components. Including achieving a full BVLOS mission profile.

Risks: Commercial risks include certification for BVLOS operations...

Defence: Kinetic threats, including autonomously aimed intercepted drones

### **PART 4 - BUSINESS CASE \* ()**

- What are the applications of your solution in the defence and security sector? Describe any routes to adoption that you have identified.
- What is the market for your solution?
- What is your current level of investment from public and/or private sources?
- Do you have existing relationships with industry primes? Do you intend to partner with a systems integrator or provide an integrated solution?

- Gate partner for Defence

## IMAGES

You may upload a maximum of two (2) images containing imagery or graphics. These images or graphics may contain words to signify elements within the images or graphs, however, they must not contain narrative text with explanations or commentary. Any images that are uploaded that attempt to supplement their proposal with additional narrative text may render the proposal non-compliant. Images of block text will be discarded.

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## Long form proposal

### PART 0 - ONE-SENTENCE DESCRIPTION \* ()

Provide an overview of your technical solution in one sentence based on the following format: "the goal is to [provide this capability] using [this technology], to achieve [this impact]."

### PART 1: ABSTRACT \* ()

Provide a short summary of your solution in one paragraph. Think of this as a written "elevator pitch".

### PART 2 - ALIGNMENT TO CHALLENGE \* ()

Describe which elements of the challenge your technical solution intends to address. Explain what is novel or unique about your technical solution.

### **PART 3 - TECHNICAL SOLUTION \* ()**

Describe your technical solution to the challenge in detail. Describe the current state of development of your solution and outline key performance metrics. Highlight any completed or planned test campaigns.

### **PART 4 - TECHNOLOGY USE CASE \* ()**

Provide example use cases of how your technical solution could be applied in defence and security settings, highlighting any direct military use cases. Describe the range of entities (military, civil protection, national security agency etc.) that could potentially adopt your technical solution. You are not expected to have a deep knowledge of defence and security operational uses but should describe some potential applications. Some examples are given in the challenge statement.

### **PART 5 - BUSINESS CASE \* ()**

- Describe the potential for the adoption of your technical solution by potential end users, and how this relates to addressable market segments and how you will exploit these. Highlight any routes to adoption that you have already identified, such as through government

contracts, defence primes, etc.

- How do you intend to scale your solution? Do you have any intellectual property rights or dependencies that impact the commercial value of the proposed solution? What is your anticipated revenue (including funding opportunities) and current financial runway? Also include any open standards that have been implemented in your solution.

#### **PART 6 - TECHNOLOGY DEVELOPMENT ROADMAP WITH RISKS AND MITIGATIONS \* ()**

- Provide an estimation of your solution's current technology readiness level ("TRL") for defence applications and, if different, for civilian applications. Describe how the DIANA programme can enable the TRL to advance and how the technical solution can progress towards a deployable capability.
- What is the overall roadmap to achieve an operational solution? What key milestones are achievable in six months? What key milestones are achievable after one year or longer? Identify key risks (technical and otherwise) associated with achieving these milestones and how you propose to mitigate these risks.

## PART 7 - TEAM AND RESOURCES \* ()

- Describe the background and experience of the personnel in your team and how they are positioned to execute the proposed roadmap and benefit from the DIANA programme. Set out how your team would continue to advance the technical solution and balance core business operations alongside the required programme engagements.
- Describe how you intend to use the €100,000 contractual challenge funding, noting that all funds must be expended in support of the DIANA programme objectives and directly related to the technical solution being proposed.

## IMAGES

You may upload a maximum of three (3) images containing imagery or graphics. These images or graphics may contain words to signify elements within the images or graphs, however, they must not contain narrative text with explanations or commentary. Any images that are uploaded that attempt to supplement their proposal with additional narrative text may render the proposal non-compliant. Images of block text will be discarded.

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## EMERGING AND DISRUPTIVE TECHNOLOGIES

Please select the NATO Emerging and Disruptive Technologies ("EDT") that apply to your technical solution. Further information can be found: [https://www.nato.int/cps/en/natohq/topics\\_184303.htm](https://www.nato.int/cps/en/natohq/topics_184303.htm)

- ☐ Artificial intelligence (AI)
- ☐ Quantum technologies

- ☐ Biotechnology and human enhancement technologies
- ☐ Space
- ☐ Hypersonic systems
- ☐ Novel materials and manufacturing
- ☐ Energy and propulsion
- ☐ Next-generation communication networks
- ☒ Autonomous Systems

#### TECHNICAL TAGS \* ()

Please select any tags that relate to your technical solution. You may select as many tags as you wish.

Select...

- AI, Autonomy & Decision Systems > Situational Awareness ✕
- Unmanned Systems (Air, Land, Sea) > Unmanned Systems ✕
- Unmanned Systems (Air, Land, Sea) > Drone UAV ✕
- Unmanned Systems (Air, Land, Sea) > UAS Uxs ✕
- Unmanned Systems (Air, Land, Sea) > Autonomous Navigation ✕
- Unmanned Systems (Air, Land, Sea) > GPS Denied Navigation ✕
- Unmanned Systems (Air, Land, Sea) > Path Planning ✕
- Unmanned Systems (Air, Land, Sea) > Obstacle Detection ✕
- Unmanned Systems (Air, Land, Sea) > SLAM Mapping ✕
- Unmanned Systems (Air, Land, Sea) > Autonomous Control ✕
- Unmanned Systems (Air, Land, Sea) > Mission Autonomy ✕
- Unmanned Systems (Air, Land, Sea) > Payload Carrier Platform ✕
- Medical, Human Factors & Physiology > Medical Monitoring ✕
- Medical, Human Factors & Physiology > Physiological Sensing ✕
- Medical, Human Factors & Physiology > Triage Systems ✕
- Medical, Human Factors & Physiology > Wound Monitoring ✕
- Medical, Human Factors & Physiology > Biomedical Carrier System ✕
- Manufacturing & Materials > Additive Manufacturing ✕
- Manufacturing & Materials > Composite Materials ✕
- Manufacturing & Materials > Fabrication Technologies ✕
- Manufacturing & Materials > Polymer Processing ✕

#### ADDITIONAL TAGS

If you wish to describe your technology with any other tags, please single tags of up to 50 characters, separated by commas.

#### AGREEMENT TO T&CS \* ()

In accordance with section 7.2(a) of the [Terms and Conditions](#).

I confirm I have read the Terms and Conditions of Bidding in full and agree to them.

#### CONFIRMATION OF COMPANY IDENTITY \* ()

This proposal is submitted by a user working for the company under which this proposal is registered; and this is the company that would enter into an agreement with DIANA if selected.

#### CONFIRMATION THAT COMPANY IS NOT A UNIVERSITY OR NON-PROFIT \* ()

I confirm that the entity submitting the proposal is not a university, non-profit or non-for-profit

organisation.

#### **PERSONNEL AGREEMENT \* ()**

I agree that the key personnel involved in the Step 3 - Virtual Panel Session, should that point be reached, will be directly involved in the accelerator programme and the required travel.

#### **ACCELERATOR PROGRAMME AGREEMENT \* ()**

I understand and accept that the accelerator programme includes approx. 6 in-person mandatory in-person thematic clusters at the assigned accelerator site and event locations involving transatlantic travel. Further I understand and accept that the accelerator programme includes mandatory completion of assignments.

## Information Sharing

#### **BIDDER IDENTITY**

In accordance with Section 7.2(a) of the Terms and Conditions of Bidding, DIANA may share bidder identity with NATO and Allied stakeholders to foster further opportunities, unless you opt out of such sharing:

#### **BIDDER DOCUMENTATION**

In accordance with Section 7.2(b) of the Terms and Conditions of Bidding, DIANA may share bidder documentation with NATO and Allied stakeholders to foster further opportunities, unless you opt out of such sharing:

## Related Company

|         |
|---------|
| DynoBay |
|---------|

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